



Fig. 10: Main screen of wireless stepper motor control

data is received, valid transmission (V_p) pin goes high to drive transistor

T5 into saturation and LED1 lights up.

An actual-size, single-side PCB for the receiver-cum-decoder circuit (Fig. 7) is shown in Fig. 8 and its component layout in Fig. 9.

Software

The software program for the user interface to control the stepper motor is written in 'C' language. The signals are generated by the `output()` function. The speed of rotation of the stepper motor can be varied by changing the argument of the `delay()` function. Direction of rotation (clockwise or

anticlockwise) depends on the switch function.

When the program (WSTEPPER.C) is loaded and run, the screen shows the welcome message. Pressing any key will lead to the main screen shown in Fig. 10. The main screen displays three messages. Pressing 'c' key rotates the stepper motor in clockwise direction, while pressing 'a' key rotates the stepper motor in anti-clockwise direction. The program can be terminated by pressing 'q' key.

EFY note. The source code and other relevant files of this article are included in this month's EFY-CD.

WSTEPPER.C

```
/*PC BASED WIRELESS STEPPER MOTOR CONTROL*/
#include<stdio.h>
#include<conio.h>
#include<dos.h>
void main()
{
    void dub(void);
    char ex[26]={ "Programmed by V.MARIYAPPAN"};
    int i;
    char ex1[22]="Programming Language: C";
    int j;
    char ex2[28]="Operating system: Windows 98";
    int k;
    char ex3[39]="PC BASED WIRELESS STEPPER MOTOR CONTROL";
    clrscr();
    for(l=0;l<39;l++)
    {
        textcolor(14);gotoxy(21+l,9);
        printf("%c",ex3[l]);
        delay(50);
    }
    sound(1900);
    delay(500);
    nosound();
    delay(100);
    for(i=0;i<26;i++)
    {
        textcolor(11);gotoxy(27+i,16);
        printf("%c",ex[i]);
        delay(30);
    }
    for(k=0;k<28;k++)
    {
        textcolor(9);gotoxy(26+k,18);
        printf("%c",ex2[k]);
        delay(30);
    }
    for(j=0;j<22;j++)
    {
        textcolor(9);gotoxy(29+j,20);
        printf("%c",ex1[j]);
        delay(30);
    }
    sound(1800);
    delay(500);
    nosound();
    getch();
    dub();
}

void dub(void)
{
    int i;
    char ex[29]="For any clarification con-
```

```
tact");
    int j;
    char ex1[31]="Email-marietech2003@yahoo.co.in";
    int k;
    void clock(void);
    void anty(void);
    char ch;
    clrscr();
    textcolor(14);gotoxy(20,6);
    printf("PC BASED WIRELESS STEPPER MOTOR CONTROL");
    textcolor(11); gotoxy(24,14);
    printf("FOR CLOCKWISE ROTATION PRESS 'c'");
    textcolor(11);gotoxy(22,16);
    printf("FOR ANTICLOCKWISE ROTATION PRESS 'a'");
    textcolor(9);gotoxy(31,20);
    printf("FOR EXIT PRESS 'q'");
    ch=getch();
    switch(ch)
    {
        case 'c':clock();break;
        case 'a':anty();break;
        case 'q':
            clrscr();
            for(j=0;j<29;j++)
            {
                textcolor(9);gotoxy(27+j,14);
                printf("%c",ex1[j]);
                delay(30);
            }
            for(k=0;k<31;k++)
            {
                textcolor(9);gotoxy(26+k,15);
                printf("%c",ex1[k]);
                delay(30);
            }
            output(0x0378,0);
            textcolor(14);gotoxy(1,25);
            printf("Press any key");
            getch();
            exit(0);
        default: clrscr();
            textcolor(12);gotoxy(33,14);
            printf("WRONG KEY PRESSED");
            for(i=0;i<5;i++)
            {
                sound(1000);
                delay(100);
                nosound();
                delay(100);
            }
            dub();
    }
}
```

```
getch();
}

void clock(void)
{
    int p=0x0378;
    char e;
    clrscr();
    textcolor(14);gotoxy(31,12);
    printf("CLOCKWISE DIRECTION");
    textcolor(9);gotoxy(1,25);
    printf("Press any key to stop");
    do
    {
        output(p,1);
        delay(200);
        output(p,2);
        delay(200);
        output(p,4);
        delay(200);
        output(p,8);
        delay(200);
        output(p,0);
        sound(1000);
        delay(200);
        nosound();
    }
    while(!kbhit());
    getch();
    dub();
}

void anty(void)
{
    int p=0x0378;
    char e;
    clrscr();
    textcolor(14);gotoxy(30,12);
    printf("ANTI CLOCKWISE DIRECTION");
    textcolor(9);gotoxy(1,25);
    printf("Press any key to stop");
    do
    {
        output(p,8);
        delay(200);
        output(p,4);
        delay(200);
        output(p,2);
        delay(200);
        output(p,1);
        delay(200);
        output(p,0);
        sound(1000);
        delay(200);
        nosound();
    }
    while(!kbhit());
    getch();
    dub();
}
```